VQMS · VENDOR QUOTATION MANAGEMENT SYSTEM

Backend API Reference

Every REST endpoint of the VQMS backend, generated directly from the source code (Spring `@RestController` classes and the FastAPI routers) on 04 November 2026. Roles are taken from `@PreAuthorize` and the security configuration of each service.

206	134	11	10
endpoints	reachable through the gateway	services with an API	gateway routes

How to call the API

- **Base URL:** clients call the **API gateway** – `http://localhost:8080` (Docker / local), or through the UI at `http://localhost:3000/api/...`. Services are not exposed directly in Docker.
- **Authentication:** `POST /api/v1/auth/login` with `{"username","password"}` returns a JWT. Send it on every other call as `Authorization: Bearer <token>`. The token carries the user id and one role.
- **Roles:** ADMIN, PROCUREMENT_ADMIN, PROCUREMENT_MANAGER, PROCUREMENT_REVIEWER, EMPLOYEE, VENDOR. INTERNAL_SERVICE is used only for service-to-service calls. "Any signed-in user" = valid token, no role check.
- **Paging:** list endpoints accept `page` (0-based), `size` and `sort=field,asc|desc` and return `{content, totalElements, totalPages, ...}`.
- **Errors:** JSON `{status, message, errors?, correlationId}`; 400 validation, 401 missing/invalid token, 403 role not allowed, 404 not found, 409/422 business rule, 503 downstream service unavailable.

GET read · POST create/action · PUT replace · PATCH change/action · DELETE remove · {id} = path parameter, `?` = optional

Services

SERVICE	MODULE	PORT	ENDPOINTS	VIA GATEWAY
Platform (Identity, Vendors, Procurement, Quotations, Settings)	`VendorQuotationApplication`	8081	124	66
Quotation Service	`quotation-service`	8082	12	0
Document Service	`document-service`	8083	10	10
AI Extraction Service (Python / FastAPI)	`ai-extraction-service`	8090	6	4

SERVICE	MODULE	PORT	ENDPOINTS	VIA GATEWAY
Validation Service	`validation-service`	8084	6	6
Comparison Service	`comparison-service`	8085	8	8
Recommendation Service	`recommendation-service`	8086	8	8
Approval Service	`approval-service`	8087	17	17
Audit Service	`audit-service`	8088	5	5
Notification Service	`notification-service`	8089	6	6
Reporting Service	`reporting-service`	8091	4	4

Gateway routing

The API gateway (`gateway-service`, port 8080) forwards requests by path prefix. Paths not listed here are not reachable from outside. Eureka (port 8761) is the service registry and has no business API.

ROUTE	PATHS	TARGET

`monolith`	`/api/v1/auth/**`
	`/api/v1/vendors/**`
	`/api/v1/procurement-requests/**`
	`/api/v1/settings/**`
	`/api/v1/quotations/**`
		`${MONOLITH_URI:http://localhost:8081}`

`document-service`	`/api/v1/documents/**`	`${DOCUMENT_SERVICE_URI:http://localhost:8083}`
`ai-extraction-service`	`/api/v1/extractions/**`	`${AI_EXTRACTION_SERVICE_URI:http://localhost:8090}`
`validation-service`	`/api/v1/validations/**`	`${VALIDATION_SERVICE_URI:http://localhost:8084}`
`comparison-service`	`/api/v1/comparisons/**`	`${COMPARISON_SERVICE_URI:http://localhost:8085}`
`recommendation-service`	`/api/v1/recommendations/**`	`${RECOMMENDATION_SERVICE_URI:http://localhost:8086}`
`approval-service`	`/api/v1/approvals/**` `/api/v1/approval-policies/**`	`${APPROVAL_SERVICE_URI:http://localhost:8087}`
`audit-service`	`/api/v1/audit/**`	`${AUDIT_SERVICE_URI:http://localhost:8088}`
`notification-service`	`/api/v1/notifications/**`	`${NOTIFICATION_SERVICE_URI:http://localhost:8089}`
`reporting-service`	`/api/v1/reports/**`	`${REPORTING_SERVICE_URI:http://localhost:8091}`

Platform (Identity, Vendors, Procurement, Quotations, Settings)

`VendorQuotationApplication` · port 8081 · 124 endpoints · 66 reachable through the gateway

> Through the gateway this application serves `/auth`, `/vendors`, `/procurement-requests`, `/quotations` and `/settings`. Its older `/approvals`, `/comparisons`, `/recommendations`, `/audit`, `/reports` and `/notifications` endpoints are shadowed by the dedicated services and are marked "internal".

ApprovalController 12 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	`/api/v1/approvals` internal – not routed by gateway	— `createApproval()`	**Body:** `ApprovalRequest`	Any signed-in user
GET	`/api/v1/approvals/approver/{approverId}` internal – not routed by gateway	— `getApprovalsByApprover()`	**Path:** approverId **Query:** page?, size?	Any signed-in user
GET	`/api/v1/approvals/overdue/{approverId}` internal – not routed by gateway	— `getOverdueApprovals()`	**Path:** approverId **Query:** page?, size?	Any signed-in user
GET	`/api/v1/approvals/pending/{approverId}` internal – not routed by gateway	— `getPendingApprovals()`	**Path:** approverId **Query:** page?, size?	Any signed-in user
GET	`/api/v1/approvals/quotation/{quotationId}/history` internal – not routed by gateway	— `getQuotationApprovalHistory()`	**Path:** quotationId	Any signed-in user
GET	`/api/v1/approvals/request/{requestId}` internal – not routed by gateway	— `getApprovalsByRequest()`	**Path:** requestId **Query:** page?, size?	Any signed-in user
GET	`/api/v1/approvals/request/{requestId}/current` internal – not routed by gateway	— `getCurrentPendingApproval()`	**Path:** requestId	Any signed-in user
GET	`/api/v1/approvals/request/{requestId}/fully-approved` internal – not routed by gateway	— `isFullyApproved()`	**Path:** requestId	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/approvals/request/{requestId}/history internal – not routed by gateway	— getApprovalHistory()	**Path:** requestId	Any signed-in user
GET	/api/v1/approvals/{id} internal – not routed by gateway	— getApproval()	**Path:** id	Any signed-in user
POST	/api/v1/approvals/{id}/decision internal – not routed by gateway	— makeDecision()	**Path:** id **Body:** `ApprovalDecisionRequest`	Any signed-in user
POST	/api/v1/approvals/{id}/delegate internal – not routed by gateway	— delegateApproval()	**Path:** id **Query:** delegatedTo	Any signed-in user

AuditController 8 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/audit internal – not routed by gateway	— getAllAudit()	**Query:** page?, size?	Any signed-in user
GET	/api/v1/audit/action/{action} internal – not routed by gateway	— getAuditByAction()	**Path:** action **Query:** page?, size?	Any signed-in user
GET	/api/v1/audit/actor/{actorId} internal – not routed by gateway	— getAuditByActor()	**Path:** actorId **Query:** page?, size?	Any signed-in user
GET	/api/v1/audit/entity/{entityType}/{entityId} internal – not routed by gateway	— getAuditByEntity()	**Path:** entityType, entityId **Query:** page?, size?	Any signed-in user
GET	/api/v1/audit/entity/{entityType}/{entityId}/history internal – not routed by gateway	— getEntityHistory()	**Path:** entityType, entityId	Any signed-in user
GET	/api/v1/audit/range internal – not routed by gateway	— getAuditByDateRange()	**Query:** start, end, page?, size?	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/audit/stats internal – not routed by gateway	— getAuditStats()	**Query:** entityType, actorId	Any signed-in user
GET	/api/v1/audit/type/{entityType} internal – not routed by gateway	— getAuditByEntityType()	**Path:** entityType **Query:** page?, size?	Any signed-in user

AuthController 8 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/auth/login	— login()	**Body:** `LoginRequest`	Public
GET	/api/v1/auth/me	— getCurrentUser()	—	Any signed-in user
PATCH	/api/v1/auth/profile	— updateProfile()	**Body:** `ProfileUpdateRequest`	Any signed-in user
POST	/api/v1/auth/register	— register()	**Body:** `RegisterRequest`	Public
GET	/api/v1/auth/users	— getAllUsers()	**Query:** page?, size?	Any signed-in user
GET	/api/v1/auth/users/{id}	— getUser()	**Path:** id	Any signed-in user
PUT	/api/v1/auth/users/{id}	— updateUser()	**Path:** id **Body:** `RegisterRequest`	Any signed-in user
PATCH	/api/v1/auth/users/{id}/deactivate	— deactivateUser()	**Path:** id	Any signed-in user

ComparisonController 8 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/comparisons/delivery/{requestId} internal – not routed by gateway	— getDeliveryComparison()	**Path:** requestId	Any signed-in user
POST	/api/v1/comparisons/generate/{requestId} internal – not routed by gateway	— generateComparison()	**Path:** requestId	Any signed-in user
GET	/api/v1/comparisons/missing-fields/{requestId} internal – not routed by gateway	— getMissingFields()	**Path:** requestId	Any signed-in user
GET	/api/v1/comparisons/price-variance/{requestId} internal – not routed by gateway	— getPriceVariance()	**Path:** requestId	Any signed-in user
GET	/api/v1/comparisons/request/{requestId} internal – not routed by gateway	— getComparisonsByRequest()	**Path:** requestId **Query:** page?, size?	Any signed-in user
GET	/api/v1/comparisons/request/{requestId}/all internal – not routed by gateway	— getAllComparisons()	**Path:** requestId	Any signed-in user
GET	/api/v1/comparisons/{id} internal – not routed by gateway	— getComparison()	**Path:** id	Any signed-in user
DELETE	/api/v1/comparisons/{id} internal – not routed by gateway	— deleteComparison()	**Path:** id	Any signed-in user

NotificationController 11 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/notifications internal – not routed by gateway	— getNotifications()	**Query:** page?, size?	Any signed-in user
POST	/api/v1/notifications internal – not routed by gateway	— sendNotification()	**Body:** `NotificationRequest`	Any signed-in user
POST	/api/v1/notifications/bulk internal – not routed by gateway	— sendBulkNotification()	**Body:** `Map<String`	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
PATCH	/api/v1/notifications/mark-all-read internal – not routed by gateway	— markAllAsRead()	—	Any signed-in user
GET	/api/v1/notifications/recent internal – not routed by gateway	— getRecentNotifications()	**Query:** limit?	Any signed-in user
GET	/api/v1/notifications/status/{status} internal – not routed by gateway	— getNotificationsByStatus()	**Path:** status **Query:** page?, size?	Any signed-in user
POST	/api/v1/notifications/template internal – not routed by gateway	— sendFromTemplate()	**Body:** `Map<String`	Any signed-in user
GET	/api/v1/notifications/type/{type} internal – not routed by gateway	— getNotificationsByType()	**Path:** type **Query:** page?, size?	Any signed-in user
GET	/api/v1/notifications/unread-count internal – not routed by gateway	— getUnreadCount()	—	Any signed-in user
DELETE	/api/v1/notifications/{id} internal – not routed by gateway	— deleteNotification()	**Path:** id	Any signed-in user
PATCH	/api/v1/notifications/{id}/read internal – not routed by gateway	— markAsRead()	**Path:** id	Any signed-in user

ProcurementRequestController 23 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/procurement-requests	— getAllRequests()	**Query:** page?, size?, sortBy?, direction?	Any signed-in user
POST	/api/v1/procurement-requests	— createRequest()	**Body:** `ProcurementRequestRequest`	Any signed-in user
GET	/api/v1/procurement-requests/my-approvals	— getRequestsForApprover()	**Query:** page?, size?, sortBy?, direction?	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/procurement-requests/my-requests	— `getMyRequests()`	**Query:** page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/procurement-requests/my-requests/search	— `searchMyRequests()`	**Query:** keyword, page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/procurement-requests/my-requests/status/{status}	— `getMyRequestsByStatus()`	**Path:** status **Query:** page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/procurement-requests/search	— `searchRequests()`	**Query:** keyword, page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/procurement-requests/status/{status}	— `getRequestsByStatus()`	**Path:** status **Query:** page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/procurement-requests/{id}	— `getRequestById()`	**Path:** id	Any signed-in user
PUT	/api/v1/procurement-requests/{id}	— `updateRequest()`	**Path:** id **Body:** `ProcurementRequestRequest`	Any signed-in user
DELETE	/api/v1/procurement-requests/{id}	— `deleteRequest()`	**Path:** id	Any signed-in user
PATCH	/api/v1/procurement-requests/{id}/approve	— `approveRequest()`	**Path:** id **Query:** comment?	Any signed-in user
PATCH	/api/v1/procurement-requests/{id}/cancel	— `cancelRequest()`	**Path:** id	Any signed-in user
PATCH	/api/v1/procurement-requests/{id}/complete	— `completeRequest()`	**Path:** id	Any signed-in user
GET	/api/v1/procurement-requests/{id}/items	— `getItems()`	**Path:** id	Any signed-in user
POST	/api/v1/procurement-requests/{id}/items	— `addItem()`	**Path:** id **Body:** `ProcurementItemRequest`	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/procurement-requests/{id}/items/{itemId}	— getItem()	**Path:** id, itemId	Any signed-in user
PUT	/api/v1/procurement-requests/{id}/items/{itemId}	— updateItem()	**Path:** id, itemId **Body:** `ProcurementItemRequest`	Any signed-in user
DELETE	/api/v1/procurement-requests/{id}/items/{itemId}	— deleteItem()	**Path:** id, itemId	Any signed-in user
PATCH	/api/v1/procurement-requests/{id}/reject	— rejectRequest()	**Path:** id **Query:** reason	Any signed-in user
PATCH	/api/v1/procurement-requests/{id}/request-quotations	— requestQuotations()	**Path:** id	Any signed-in user
PATCH	/api/v1/procurement-requests/{id}/start-evaluation	— startEvaluation()	**Path:** id	Any signed-in user
PATCH	/api/v1/procurement-requests/{id}/submit	— submitRequest()	**Path:** id	Any signed-in user

QuotationController 19 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/quotations	— getAllQuotations()	—	Any signed-in user
POST	/api/v1/quotations/manual	— createQuotation()	**Body:** `QuotationRequest`	Any signed-in user
GET	/api/v1/quotations/request/{requestId}	— getQuotationsByRequest()	**Path:** requestId **Query:** page?, size?	Any signed-in user
GET	/api/v1/quotations/vendor/{vendorId}	— getQuotationsByVendor()	**Path:** vendorId **Query:** page?, size?	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/quotations/{id}	— getQuotation()	**Path:** id	Any signed-in user
PUT	/api/v1/quotations/{id}	— updateQuotation()	**Path:** id **Body:** `QuotationRequest`	Any signed-in user
DELETE	/api/v1/quotations/{id}	— deleteQuotation()	**Path:** id	Any signed-in user
POST	/api/v1/quotations/{id}/cancel	— cancelQuotation()	**Path:** id	Any signed-in user
GET	/api/v1/quotations/{id}/items	— getItems()	**Path:** id	Any signed-in user
POST	/api/v1/quotations/{id}/items	— addItem()	**Path:** id **Body:** `QuotationItemRequest`	Any signed-in user
POST	/api/v1/quotations/{id}/process	— processQuotation()	**Path:** id	Any signed-in user
POST	/api/v1/quotations/{id}/retry	— retryProcessing()	**Path:** id	Any signed-in user
POST	/api/v1/quotations/{id}/review	— reviewQuotation()	**Path:** id	Any signed-in user
POST	/api/v1/quotations/{id}/submit	— submitQuotation()	**Path:** id	Any signed-in user
POST	/api/v1/quotations/{id}/version	— createNewVersion()	**Path:** id **Body:** `QuotationRequest`	Any signed-in user
PUT	/api/v1/quotations/{quotationId}/items/{itemId}	— updateItem()	**Path:** quotationId, itemId **Body:** `QuotationItemRequest`	Any signed-in user
DELETE	/api/v1/quotations/{quotationId}/items/{itemId}	— deleteItem()	**Path:** quotationId, itemId	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/quotations/{quotationId}/items/{itemId}/verify	— verifyItem()	**Path:** quotationId, itemId	Any signed-in user
POST	/api/v1/quotations/{requestId}/upload	— uploadQuotation()	**Path:** requestId **Query:** vendorId, file **Body:** `multipart/form-data`	Any signed-in user

RecommendationController 7 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/recommendations/generate-with-criteria/{requestId} internal – not routed by gateway	— generateRecommendationWithCriteria()	**Path:** requestId **Body:** `RecommendationCriteriaRequest`	Any signed-in user
POST	/api/v1/recommendations/generate/{requestId} internal – not routed by gateway	— generateRecommendation()	**Path:** requestId	Any signed-in user
GET	/api/v1/recommendations/latest/{requestId} internal – not routed by gateway	— getLatest()	**Path:** requestId	Any signed-in user
GET	/api/v1/recommendations/request/{requestId} internal – not routed by gateway	— getRecommendations()	**Path:** requestId **Query:** page?, size?	Any signed-in user
GET	/api/v1/recommendations/request/{requestId}/all internal – not routed by gateway	— getAllRecommendations()	**Path:** requestId	Any signed-in user
GET	/api/v1/recommendations/{id} internal – not routed by gateway	— getRecommendation()	**Path:** id	Any signed-in user
DELETE	/api/v1/recommendations/{id} internal – not routed by gateway	— deleteRecommendation()	**Path:** id	Any signed-in user

ReportingController 12 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/reports internal – not routed by gateway	— getReports()	**Query:** page?, size?	Any signed-in user
GET	/api/v1/reports/analytics internal – not routed by gateway	— getAnalytics()	—	Any signed-in user
GET	/api/v1/reports/approval/summary internal – not routed by gateway	— getApprovalSummary()	—	Any signed-in user
GET	/api/v1/reports/cost-saving internal – not routed by gateway	— getCostSaving()	—	Any signed-in user
GET	/api/v1/reports/dashboard internal – not routed by gateway	— getDashboardStats()	—	Any signed-in user
POST	/api/v1/reports/generate internal – not routed by gateway	— generateReport()	**Body:** `Map<String`	Any signed-in user
GET	/api/v1/reports/procurement/range internal – not routed by gateway	— getProcurementByDateRange()	—	Any signed-in user
GET	/api/v1/reports/procurement/summary internal – not routed by gateway	— getProcurementSummary()	—	Any signed-in user
GET	/api/v1/reports/quotation/range internal – not routed by gateway	— getQuotationByDateRange()	—	Any signed-in user
GET	/api/v1/reports/quotation/summary internal – not routed by gateway	— getQuotationSummary()	—	Any signed-in user
GET	/api/v1/reports/type/{type} internal – not routed by gateway	— getReportsByType()	**Path:** type **Query:** page?, size?	Any signed-in user
GET	/api/v1/reports/vendor/performance internal – not routed by gateway	— getVendorPerformance()	—	Any signed-in user

SystemSettingController 5 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/settings	— `getAllSettings()`	**Query:** page?, size?	Any signed-in user
POST	/api/v1/settings	— `createSetting()`	**Body:** `SystemSettingRequest`	Any signed-in user
GET	/api/v1/settings/{key}	— `getSetting()`	**Path:** key	Any signed-in user
PUT	/api/v1/settings/{key}	— `updateSetting()`	**Path:** key **Body:** `SystemSettingRequest`	Any signed-in user
DELETE	/api/v1/settings/{key}	— `deleteSetting()`	**Path:** key	Any signed-in user

VendorController 11 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/vendors	— `getAllVendors()`	**Query:** page?, size?, sortBy?, direction?	Any signed-in user
POST	/api/v1/vendors	— `createVendor()`	**Body:** `VendorRequest`	Any signed-in user
GET	/api/v1/vendors/page	— `getVendorsWithPagination()`	**Query:** page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/vendors/search	— `searchVendors()`	**Query:** keyword?, page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/vendors/status/{status}	— `getVendorsByStatus()`	**Path:** status **Query:** page?, size?, sortBy?, direction?	Any signed-in user
GET	/api/v1/vendors/{id}	— `getVendorById()`	**Path:** id	Any signed-in user

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
PUT	/api/v1/vendors/{id}	— updateVendor()	**Path:** id **Body:** `VendorRequest`	Any signed-in user
DELETE	/api/v1/vendors/{id}	— deleteVendor()	**Path:** id	Any signed-in user
PATCH	/api/v1/vendors/{id}/activate	— activateVendor()	**Path:** id	Any signed-in user
GET	/api/v1/vendors/{id}/active	— isVendorActive()	**Path:** id	Any signed-in user
PATCH	/api/v1/vendors/{id}/deactivate	— deactivateVendor()	**Path:** id	Any signed-in user

Quotation Service

`quotation-service` · port 8082 · 12 endpoints · 0 reachable through the gateway

> Not reachable through the API gateway: the gateway sends `/api/v1/quotations/**` to the Platform application. This service also still uses temporary HTTP Basic authentication.

QuotationController 12 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	`/api/v1/quotations` internal – not routed by gateway	— `createQuotation()`	**Body:** `QuotationRequest`	VENDOR
GET	`/api/v1/quotations/my-quotations` internal – not routed by gateway	— `getMyQuotations()`	**Query:** page?, size?, sortBy?	VENDOR
GET	`/api/v1/quotations/procurement-request/{procurementRequestId}` internal – not routed by gateway	— `getProcurementQuotations()`	**Path:** procurementRequestId **Query:** page?, size?, sortBy?	PROCUREMENT_ADMIN PROCUREMENT_REVIEWER
GET	`/api/v1/quotations/status/{status}` internal – not routed by gateway	— `getQuotationsByStatus()`	**Path:** status **Query:** page?, size?	PROCUREMENT_ADMIN PROCUREMENT_REVIEWER
GET	`/api/v1/quotations/{quotationId}` internal – not routed by gateway	— `getQuotationById()`	**Path:** quotationId	VENDOR PROCUREMENT_ADMIN PROCUREMENT_REVIEWER
PUT	`/api/v1/quotations/{quotationId}` internal – not routed by gateway	— `updateQuotation()`	**Path:** quotationId **Body:** `QuotationRequest`	VENDOR
DELETE	`/api/v1/quotations/{quotationId}` internal – not routed by gateway	— `deleteDraftQuotation()`	**Path:** quotationId	VENDOR
PATCH	`/api/v1/quotations/{quotationId}/accept` internal – not routed by gateway	— `acceptQuotation()`	**Path:** quotationId	PROCUREMENT_ADMIN
PATCH	`/api/v1/quotations/{quotationId}/reject` internal – not routed by gateway	— `rejectQuotation()`	**Path:** quotationId	PROCUREMENT_ADMIN PROCUREMENT_REVIEWER

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
PATCH	/api/v1/quotations/{quotationId}/review internal – not routed by gateway	— markUnderReview()	**Path:** quotationId	PROCUREMENT_ADMIN PROCUREMENT_REVIEWER
PATCH	/api/v1/quotations/{quotationId}/submit internal – not routed by gateway	— submitQuotation()	**Path:** quotationId	VENDOR
PATCH	/api/v1/quotations/{quotationId}/withdraw internal – not routed by gateway	— withdrawQuotation()	**Path:** quotationId	VENDOR

Document Service

`document-service` · port 8083 · 10 endpoints · 10 reachable through the gateway

DocumentController 10 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/documents	Search documents (vendors only see their own) `search()`	**Query:** (filter object), page?, size?, sort?	Any signed-in user
GET	/api/v1/documents/procurement-request/{requestId}	List documents of a procurement request `byProcurementRequest()`	**Path:** requestId **Query:** (filter object), page?, size?, sort?	Any signed-in user
GET	/api/v1/documents/quotation/{quotationId}	List documents of a quotation `byQuotation()`	**Path:** quotationId **Query:** (filter object), page?, size?, sort?	Any signed-in user
POST	/api/v1/documents/upload	Upload a document `upload()`	**Query:** (filter object) **Body:** `multipart/form-data`	VENDOR ADMIN PROCUREMENT_ADMIN
GET	/api/v1/documents/vendor/{vendorId}	List documents of a vendor `byVendor()`	**Path:** vendorId **Query:** (filter object), page?, size?, sort?	Any signed-in user
GET	/api/v1/documents/{id}	Get document metadata `getById()`	**Path:** id	Any signed-in user
DELETE	/api/v1/documents/{id}	Soft delete a document `delete()`	**Path:** id	VENDOR ADMIN PROCUREMENT_ADMIN
GET	/api/v1/documents/{id}/download	Get a short-lived download link `download()`	**Path:** id	Any signed-in user
GET	/api/v1/documents/{id}/history	Get document status history `history()`	**Path:** id	Any signed-in user
PATCH	/api/v1/documents/{id}/status	Change document status `updateStatus()`	**Path:** id **Body:** `DocumentStatusUpdateRequest`	ADMIN PROCUREMENT_ADMIN INTERNAL_SERVICE

AI Extraction Service (Python / FastAPI)

`ai-extraction-service` · port 8090 · 6 endpoints · 4 reachable through the gateway

extraction_routes.py 4 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/extractions	Request extraction of a quotation document `create_extraction()`	**Body:** `ExtractionRequest`	ADMIN PROCUREMENT_ADMIN INTERNAL_SERVICE
GET	/api/v1/extractions/document/{document_id}	List extractions of a document (newest first) `list_document_extractions()`	**Path:** document_id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
GET	/api/v1/extractions/{extraction_id}	Get extraction status and result `get_extraction()`	**Path:** extraction_id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	/api/v1/extractions/{extraction_id}/retry	Retry a failed extraction `retry_extraction()`	**Path:** extraction_id	ADMIN PROCUREMENT_ADMIN INTERNAL_SERVICE

health_routes.py 2 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/health internal – not routed by gateway	Liveness: the process is running `liveness()`	—	Public
GET	/api/v1/health/ready internal – not routed by gateway	Readiness: database and message broker reachable `readiness()`	—	Public

Validation Service

`validation-service` · port 8084 · 6 endpoints · 6 reachable through the gateway

ValidationController 6 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/validations	Search validations `search()`	**Query:** status?, vendorId?, procurementRequestId?, (filter object), page?, size?, sort?	Any signed-in user
POST	/api/v1/validations	Validate a quotation (optionally with its AI extraction) `validate()`	**Body:** `ValidationRequest`	ADMIN PROCUREMENT_ADMIN INTERNAL_SERVICE
GET	/api/v1/validations/quotation/{quotationId}	Validation history of a quotation (newest first) `byQuotation()`	**Path:** quotationId **Query:** (filter object), page?, size?, sort?	Any signed-in user
GET	/api/v1/validations/status/{status}	Validations by status `byStatus()`	**Path:** status **Query:** (filter object), page?, size?, sort?	Any signed-in user
GET	/api/v1/validations/{id}	Get a validation report with all issues `getById()`	**Path:** id	Any signed-in user
POST	/api/v1/validations/{id}/revalidate	Run validation again (creates a new record linked to the previous one) `revalidate()`	**Path:** id	ADMIN PROCUREMENT_ADMIN

Comparison Service

`comparison-service` · port 8085 · 8 endpoints · 8 reachable through the gateway

ComparisonController 8 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/comparisons	Search comparisons `search()`	**Query:** status?, procurementRequestId?, (filter object), page?, size?, sort?	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	/api/v1/comparisons	Create a comparison with procurement-specific criteria weights `create()`	**Body:** `ComparisonRequest`	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
GET	/api/v1/comparisons/procurement-request/{id}	Comparisons of a procurement request `byProcurementRequest()`	**Path:** id **Query:** (filter object), page?, size?, sort?	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
GET	/api/v1/comparisons/{id}	Get a comparison `getById()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	/api/v1/comparisons/{id}/calculate	Calculate (or recalculate) scores and ranking `calculate()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
POST	/api/v1/comparisons/{id}/complete	Complete the comparison and send it to the Recommendation Service `complete()`	**Path:** id	ADMIN PROCUREMENT_ADMIN
GET	/api/v1/comparisons/{id}/ranking	Ranking of eligible quotations `ranking()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/comparisons/{id}/results	Detailed comparison report results()	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE

Recommendation Service

`recommendation-service` · port 8086 · 8 endpoints · 8 reachable through the gateway

RecommendationController 8 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/recommendations	Search recommendations `search()`	**Query:** status?, procurementRequestId?, (filter object), page?, size?, sort?	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	/api/v1/recommendations	Generate a recommendation for a completed comparison `generate()`	**Body:** `RecommendationRequest`	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
GET	/api/v1/recommendations/procurement-request/{id}	Recommendation history of a procurement request `byProcurementRequest()`	**Path:** id **Query:** (filter object), page?, size?, sort?	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
GET	/api/v1/recommendations/{id}	Get a recommendation `getById()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
GET	/api/v1/recommendations/{id}/explanation	Explanation of the recommendation `explanation()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
GET	/api/v1/recommendations/{id}/ranking	Ranking with scores and risk `ranking()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	/api/v1/recommendations/{id}/regenerate	Regenerate (supersedes the current recommendation) `regenerate()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/recommendations/{id}/submit-for-approval	Submit to the Approval Service (does not approve anything) submitForApproval()	**Path:** id **Body:** SubmitForApprovalRequest	ADMIN PROCUREMENT_ADMIN

Approval Service

`approval-service` · port 8087 · 17 endpoints · 17 reachable through the gateway

ApprovalPolicyController 5 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	`/api/v1/approval-policies`	List approval policies `list()`	—	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
POST	`/api/v1/approval-policies`	Create an approval policy `create()`	**Body:** `ApprovalPolicyRequest`	ADMIN
GET	`/api/v1/approval-policies/{id}`	Get an approval policy `get()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
PUT	`/api/v1/approval-policies/{id}`	Update an approval policy (applies to new approvals only) `update()`	**Path:** id **Body:** `ApprovalPolicyRequest`	ADMIN
PATCH	`/api/v1/approval-policies/{id}/deactivate`	Deactivate an approval policy `deactivate()`	**Path:** id	ADMIN

ApprovalController 12 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	`/api/v1/approvals`	Search approvals `search()`	**Query:** status?, procurementRequestId?, (filter object), page?, size?, sort?	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	`/api/v1/approvals`	Open an approval for a recommendation submitted for approval `create()`	**Body:** `ApprovalRequest`	ADMIN PROCUREMENT_ADMIN INTERNAL_SERVICE

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/approvals/delegations	Delegate your approval authority for a period delegate()	**Body:** ApprovalDelegationRequest	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE
GET	/api/v1/approvals/delegations/my	Delegations given or received by the current user myDelegations()	—	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
DELETE	/api/v1/approvals/delegations/{delegationId}	Revoke a delegation revoke()	**Path:** delegationId	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
GET	/api/v1/approvals/pending	Approvals awaiting the current user's decision pending()	—	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
GET	/api/v1/approvals/{id}	Get an approval with its levels getById()	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	/api/v1/approvals/{id}/approve	Approve the pending level the caller is authorised for approve()	**Path:** id **Body:** ApprovalDecisionRequest	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE
POST	/api/v1/approvals/{id}/cancel	Cancel an open approval (reason required) cancel()	**Path:** id **Body:** ApprovalDecisionRequest	ADMIN PROCUREMENT_ADMIN
POST	/api/v1/approvals/{id}/escalate	Escalate pending levels manually escalate()	**Path:** id **Body:** ApprovalDecisionRequest	ADMIN PROCUREMENT_ADMIN
GET	/api/v1/approvals/{id}/history	Audit trail of an approval history()	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/approvals/{id}/reject	Reject (comment required) reject()	**Path:** id **Body:** ApprovalDecisionRequest	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER EMPLOYEE INTERNAL_SERVICE

Audit Service

`audit-service` · port 8088 · 5 endpoints · 5 reachable through the gateway

AuditController 5 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/audit/aggregates/{aggregateType}/{aggregateId}	Chronological history of one business object `timeline()`	**Path:** aggregateType, aggregateId	ADMIN PROCUREMENT_ADMIN
GET	/api/v1/audit/events	Search the audit trail (newest first) `search()`	**Query:** sourceService?, eventType?, aggregateType?, aggregateId?, actorId?, (filter object), page?, size?, sort?	ADMIN PROCUREMENT_ADMIN
POST	/api/v1/audit/events	Record an audit event (idempotent per eventId) `record()`	**Body:** `AuditEventRequest`	INTERNAL_SERVICE ADMIN
GET	/api/v1/audit/events/{id}	Get one audit event `get()`	**Path:** id	ADMIN PROCUREMENT_ADMIN
GET	/api/v1/audit/verify	Verify the integrity of the whole hash chain `verify()`	—	ADMIN

Notification Service

`notification-service` · port 8089 · 6 endpoints · 6 reachable through the gateway

NotificationController 6 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
POST	/api/v1/notifications	Send a notification to a user or to all users of a role `send()`	**Body:** `NotificationRequest`	INTERNAL_SERVICE, ADMIN, PROCUREMENT_ADMIN
GET	/api/v1/notifications/my	My notifications (newest first) `my()`	**Query:** unreadOnly?, (filter object), page?, size?, sort?	Any signed-in user
PATCH	/api/v1/notifications/my/read-all	Mark all my notifications as read `markAllRead()`	—	Any signed-in user
GET	/api/v1/notifications/my/unread-count	Number of unread notifications `unreadCount()`	—	Any signed-in user
GET	/api/v1/notifications/{id}	Get one of my notifications `get()`	**Path:** id	Any signed-in user
PATCH	/api/v1/notifications/{id}/read	Mark a notification as read `markRead()`	**Path:** id	Any signed-in user

Reporting Service

`reporting-service` · port 8091 · 4 endpoints · 4 reachable through the gateway

ReportController 4 endpoints

METHOD	PATH	DESCRIPTION	PARAMETERS	ALLOWED ROLES
GET	/api/v1/reports	List generated reports (newest first) `list()`	**Query:** type?, (filter object), page?, size?, sort?	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
POST	/api/v1/reports	Generate a report (stored as a snapshot) `generate()`	**Body:** `ReportRequest`	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
GET	/api/v1/reports/{id}	Get a stored report with its data `get()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER
GET	/api/v1/reports/{id}/export	Export a stored report as CSV `export()`	**Path:** id	ADMIN PROCUREMENT_ADMIN PROCUREMENT_MANAGER